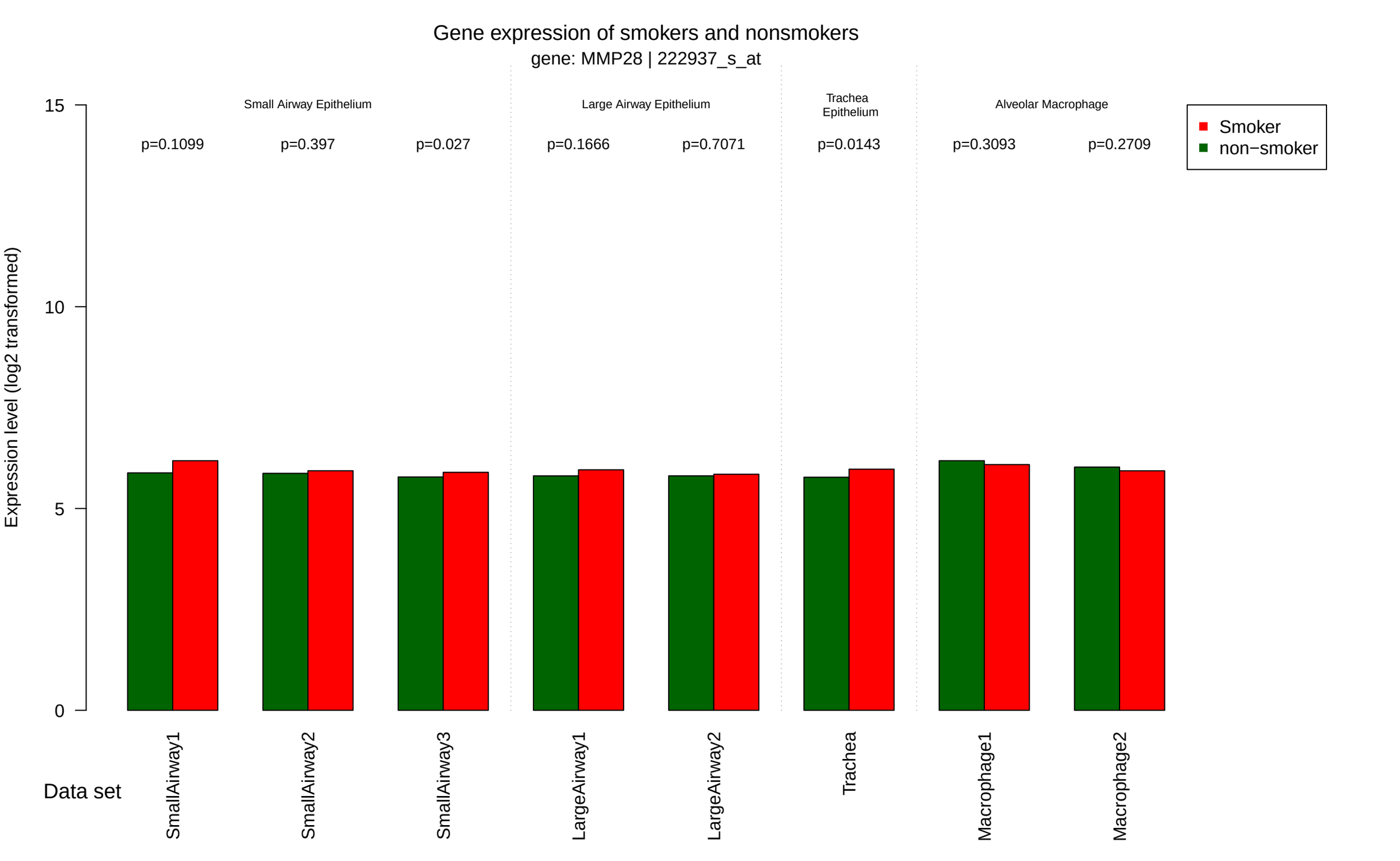
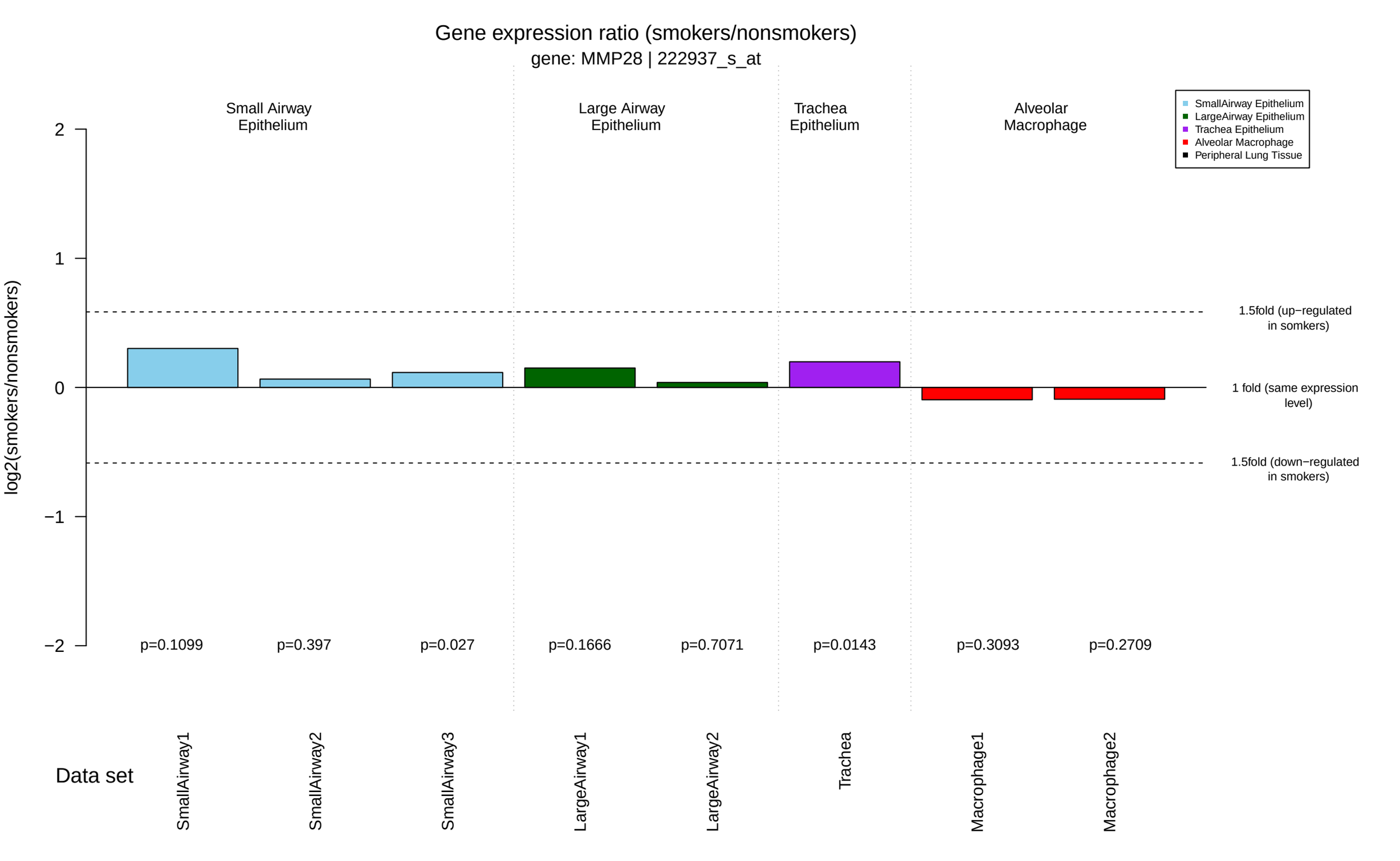
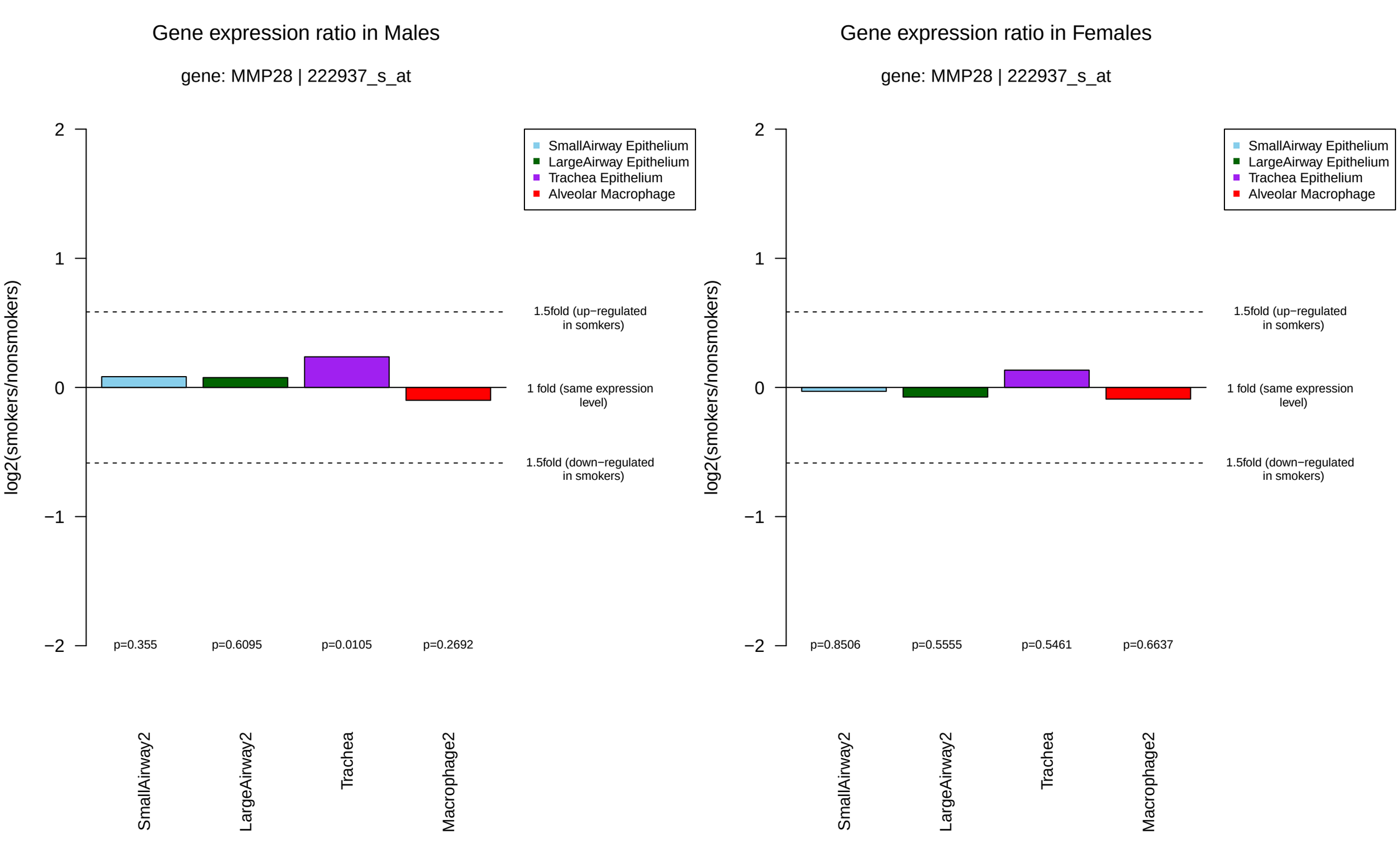
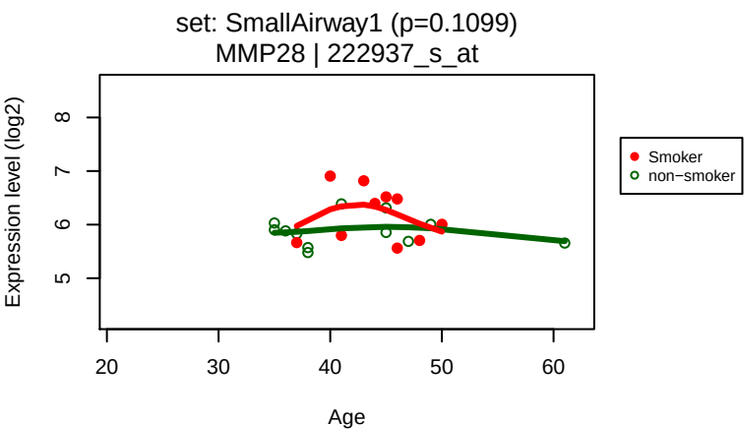
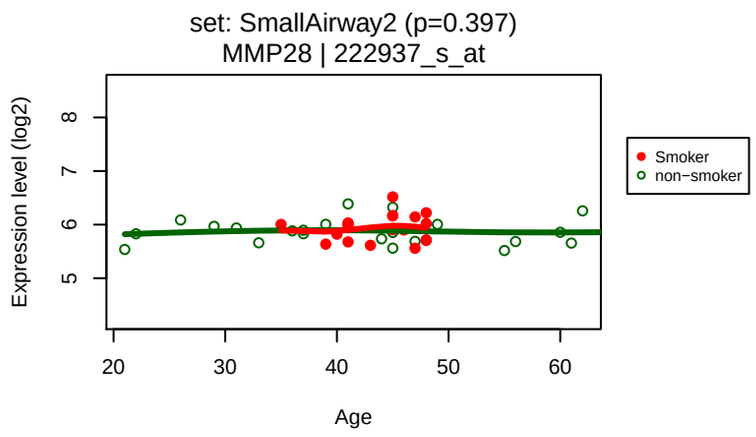
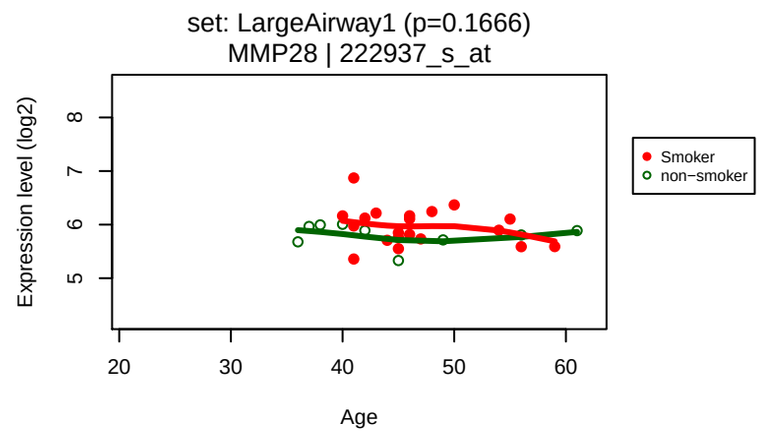

Gene expression of smokers and nonsmokers
gene: MMP28 | 222937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1099
p=0.397
p=0.027
p=0.1666
p=0.7071
p=0.0143
p=0.3093
p=0.2709
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

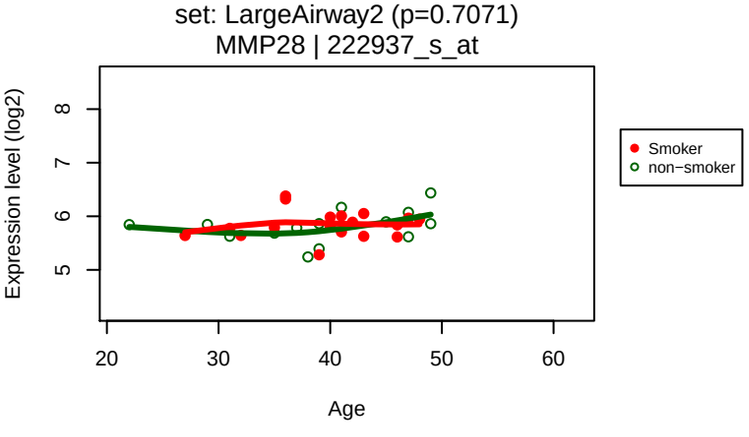

Gene expression ratio (smokers/nonsmokers)
gene: MMP28 | 222937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1099
p=0.397
p=0.027
p=0.1666
p=0.7071
p=0.0143
p=0.3093
p=0.2709
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

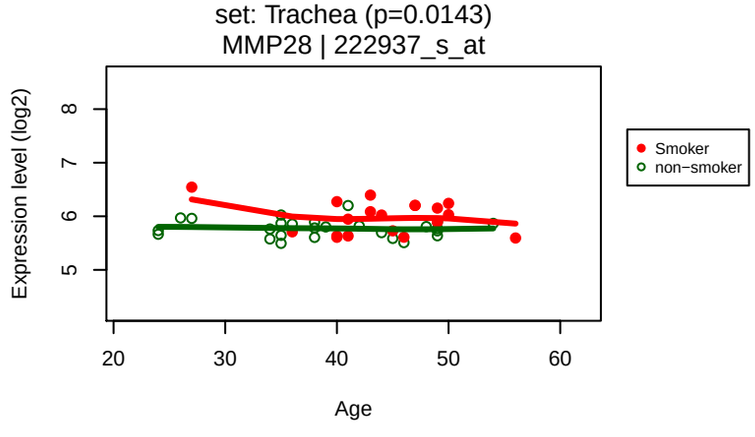

Gene expression ratio in Males
gene: MMP28 | 222937_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.355
p=0.6095
p=0.0105
p=0.2692
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP28 | 222937_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8506
p=0.5555
p=0.5461
p=0.6637
SmallAirway2
LargeAirway2
Trachea
Macrophage2

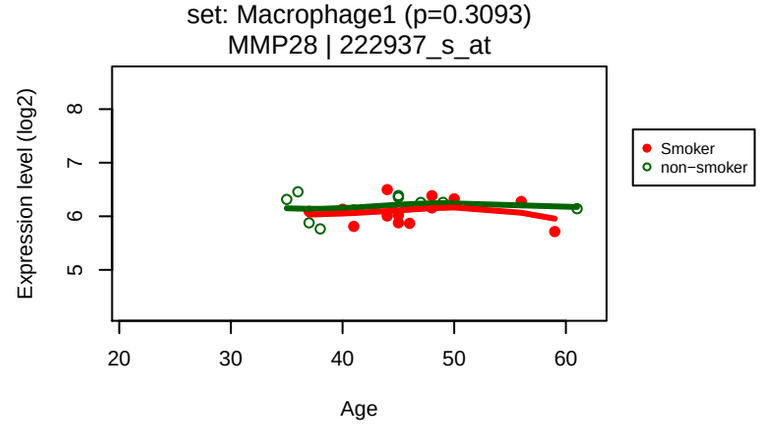

set: SmallAirway1 (p=0.1099)
MMP28 | 222937_s_at
Expression level (log2)
Age
Smoker
non−smoker

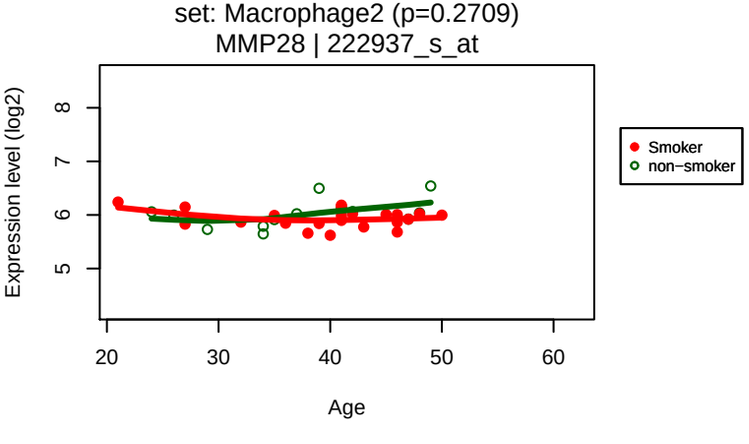

set: SmallAirway2 (p=0.397)
MMP28 | 222937_s_at
Expression level (log2)
Age
Smoker
non−smoker

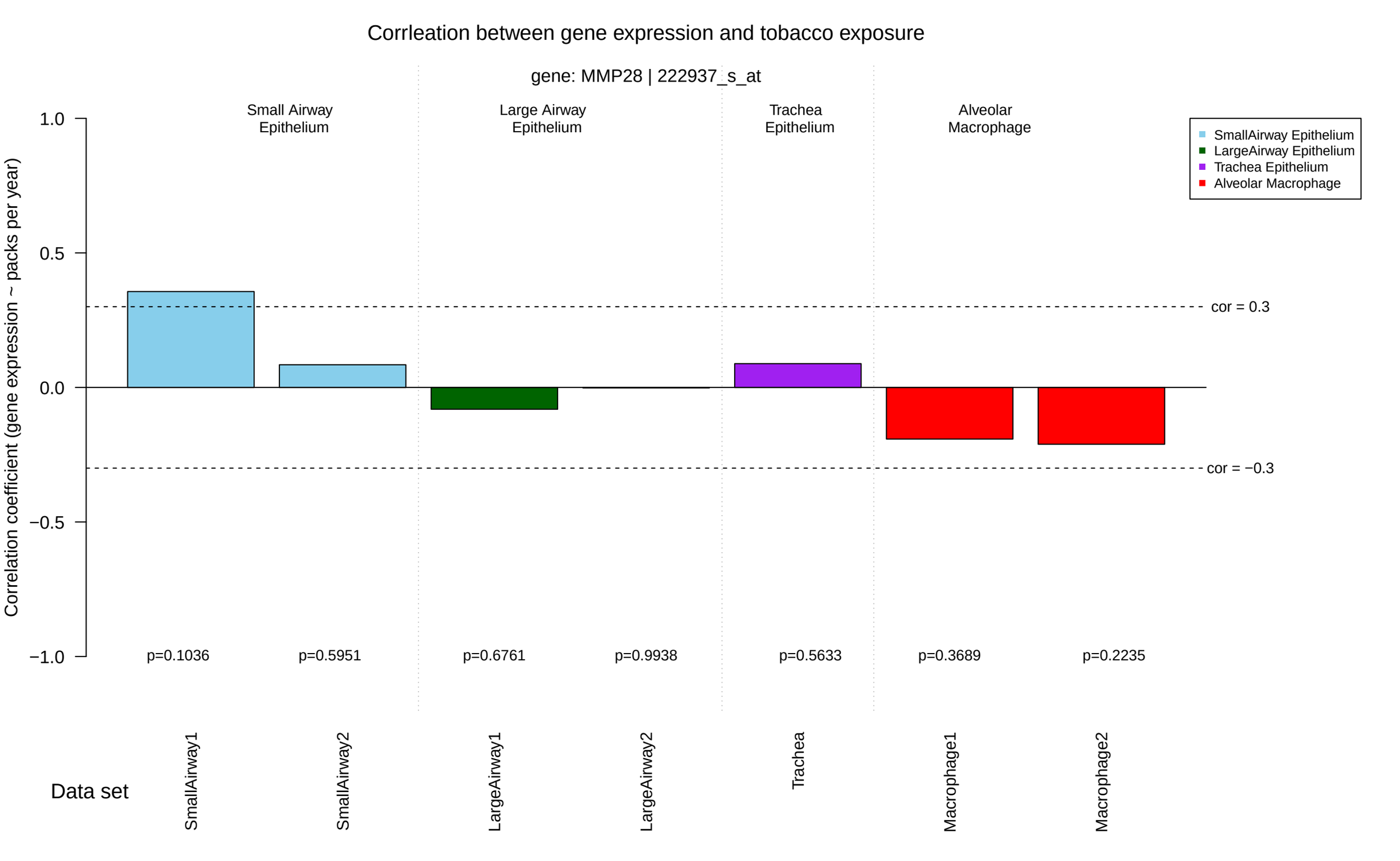

set: LargeAirway1 (p=0.1666)
MMP28 | 222937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7071)
MMP28 | 222937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0143)
MMP28 | 222937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3093)
MMP28 | 222937_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2709)
MMP28 | 222937_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP28 | 222937_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1036
p=0.5951
p=0.6761
p=0.9938
p=0.5633
p=0.3689
p=0.2235
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2